Homework III

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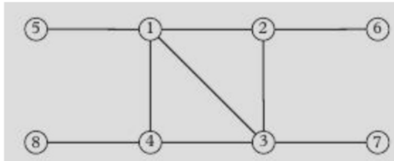
Date: 23 November, 2025

1. Consider the network shown in the figure.



a) The average shortest path length L for a network of N nodes is defined by

$$L = \frac{1}{N(N-1)} \sum_{i \neq j} d_{ij},$$

where d_{ij} is the length of the shortest path between node i and node j.
Calculate L for this network. [1]

$$\sum_{i \neq j} d_{ij} = \sum_{j} \left(d_{1j} + d_{2j} + d_{3j} + d_{4j} + d_{5j} + d_{6j} + d_{7j} + d_{8j} \right)$$

For the node $i=1$:

$$\sum_j d_{1j} = d_{12} + d_{13} + d_{14} + d_{15} + d_{16} + d_{17} + d_{18} = 10$$

For the node $i=2$:

$$\sum_j d_{2j} = 12$$

For the node $i=3$

$$\sum_j d_{3j} = 10$$

For the node $i=4$:

$$\sum_j d_{4j} = 12$$

For the node $i=5$:

$$\sum_j d_{5j} = 16$$

For the node $i=6$

$$\sum_j d_{6j} = 18$$

For the node $i=7$

$$\sum_j d_{7j} = 16$$

For the node $i=8$

$$\sum_j d_{8j} = 18$$

$$\Rightarrow L = \frac{1}{8(8-1)} \left(10 + 12 + 10 + 12 + 16 + 18 + 16 + 18 \right)$$

$$L = \frac{112}{56} = 2$$

$$\boxed{L = 2}$$

b) The local clustering coefficient of a node i is defined as

$$C(i) = \frac{2t_i}{k_i(k_i - 1)},$$

where t_i is the number of triangles containing node i and k_i is the number of neighbors of node i.
The clustering coefficient of the network is the average

$$\langle C \rangle = \frac{1}{N} \sum_{i=1}^{N} C(i).$$

Calculate the average clustering coefficient $\langle C \rangle$ for this network. [1]

Node	k_i	t_i	$C(i)$
1	4	1	2/12
2	3	1	2/6
3	4	1	2/12
4	3	1	2/6
5	1	0	0
6	1	0	0
7	1	0	0
8	1	0	0

$$\Rightarrow \langle C \rangle = \frac{1}{8}\left(\frac{2}{12} + \frac{2}{6} + \frac{2}{12} + \frac{2}{6}\right)$$

$$= \frac{1}{8}\left(\frac{4}{6} + \frac{4}{12}\right) = \frac{1}{8}\left(\frac{12}{12}\right) = \frac{1}{8}$$

$$\boxed{\langle C \rangle = 0.125}$$

c) The degree or number of neighbors of node i is denoted by k_i. Calculate the average degree of this network. [1]

$$\langle K \rangle = \frac{1}{N} \sum k_i$$

$$= \frac{1}{8}(4 + 3 + 4 + 3 + 4)$$

$$= \frac{18}{8}$$

$$\boxed{\langle K \rangle = 2.25}$$

2. Describe the properties of the following networks and the mechanisms proposed for their formation:
   a) Small world networks. [1]
   b) Scale-free networks. [1]
   c) Communities. [1]

a) Small world networks

Properties:

- These networks exist between the order and disorder.
- They have long-range connections.
- Short average path length (L): Any two nodes can be reached in few steps.

Formation mechanism:

1. Start with a regular ring lattice.
2. Rewire a small fraction of edges (randomly).

b) Scale-free networks

Properties:

- Power law distribution:

$$P(K) \propto K^{-\alpha}$$

  This means that few nodes have a lot of connections while most of the nodes have few connections.
- Hub dominance: Highly connected nodes control much of the network structure.

Formation mechanism:

- The network expands by adding new nodes.
- Preferential connections: New nodes prefer nodes with a large amount of connections.

c) Communities

Properties:

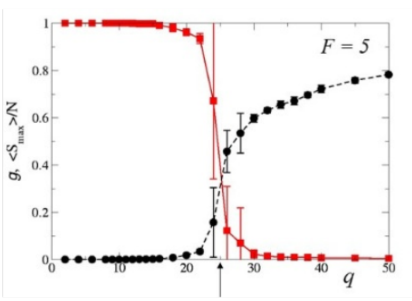
- Nodes form modules (communities) with dense internal connections and sparse connections between different communities.
- This could represent social groups (friends networks).

Formation mechanism:

- Homophily: Nodes connect to other nodes that are similar which has same function, interest or attributes.
- Nodes tend to connect to neighbors-of-neighbors, reforcing the formation of communities.

3. a) Describe the phase transition found in the Axelrod model for cultural influence. [1]
   b) Describe the counterintuitive effect of external mass media in Axelrod model. [1]

a)



$q_c \approx 25$

First, it is important to remark that q depends on F and it is the number of equivalent cultural states.

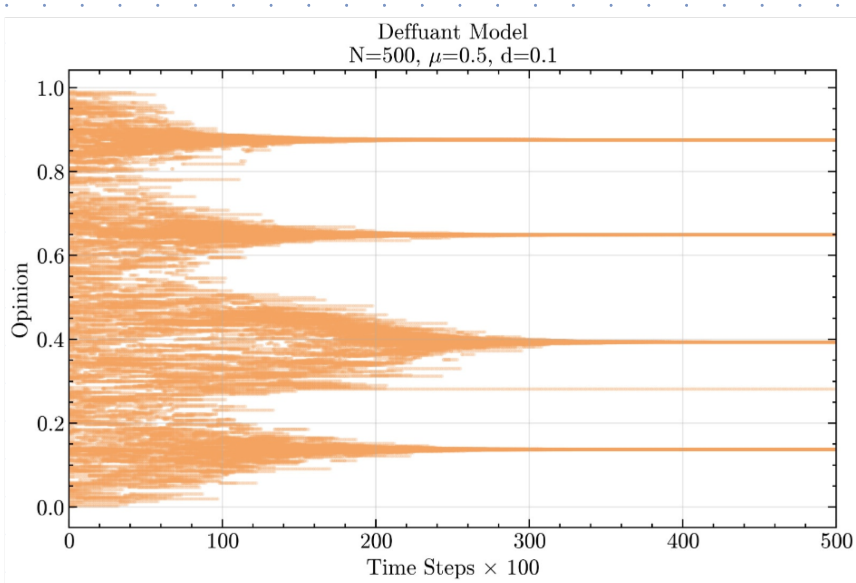
- The phase transition that we observe in this model ($q_c \approx 25$) referes to the diversity when we vary the number of options availables.
- g is the mean fraction of number of domains and it increase when $q > q_c$, and decrease on the other side which referes to the cultural diversity and cultural homogeneity, respectively.
- As well as g, $\langle S_{max} \rangle / N$ shows how the number of options affects the cultural diversity. In short, when we have more options it is possible to develop diversity. From our physical point of view diversity is a phase transition of the same system.

b)

The critical value q_c decreases when the intensity of the external mass media increases. This means that this external "field" is inducing disorder (more diversity). It is seen that there is an threshold value where the intesity of the field for a specific q does not convince anymore. However, this field induce heterogeneity. We can relate this with advertising: low advertising may be more effective to convince more people.

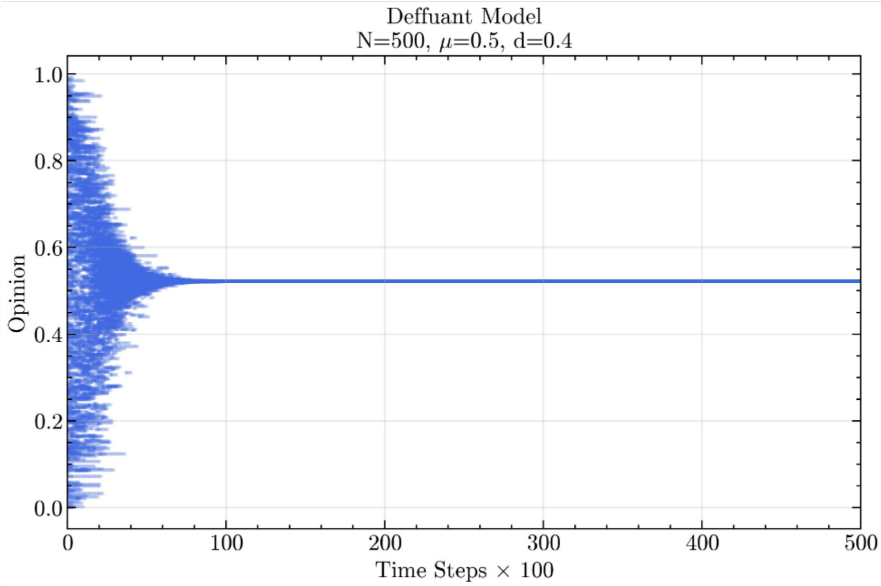
4. Implement a computer code for Deffuant's bounded confidence model in a globally coupled network of N agents with approaching parameter μ and interaction threshold d (see Class Notes). You may use the programming language of your choice. Plot the temporal evolution of the opinion states of the $N = 500$ agents until clusters are formed, with fixed parameters:
   a) $\mu = 0.5, d = 0.1$. [1]
   b) $\mu = 0.5, d = 0.4$. [1]

a)



With this conditions we see the formation of 5 clusters after 30000 time steps. Since d is very short, agents only interact if their opinion differ by < 0.1. This leads to the formation of small clusters.

b)



With this conditions we see the formation of 1 cluster after 10000 time steps. The system reach full "consensus" since $d=0.4$ is a large confidence interval.